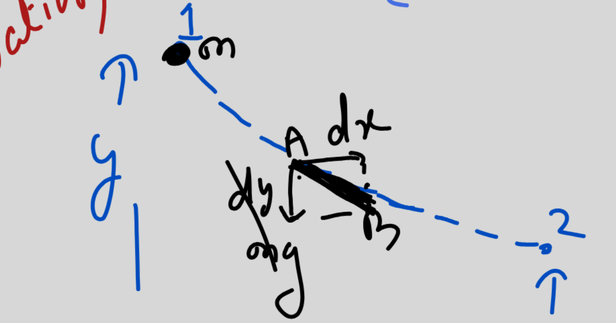
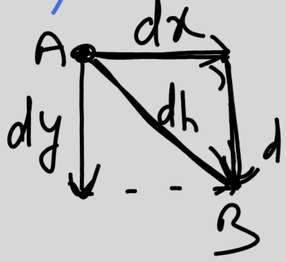
P.t Gravitational Force/field is conservative in nature:

Imp Derivation





$\vec{dh} = \vec{dx} + \vec{dy}$

$= dx\hat{i} + dy(-\hat{j})$

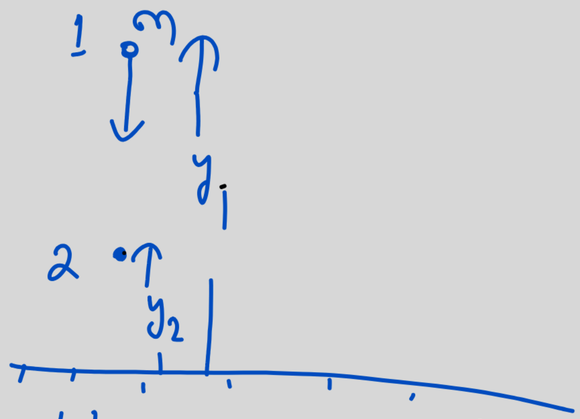
$\Rightarrow \vec{dh} = dx\hat{i} - dy\hat{j}$

$\vec{F}_{grav} = mg(-\hat{j})$

$dw = \vec{F}_{grav} \cdot \vec{dh}$

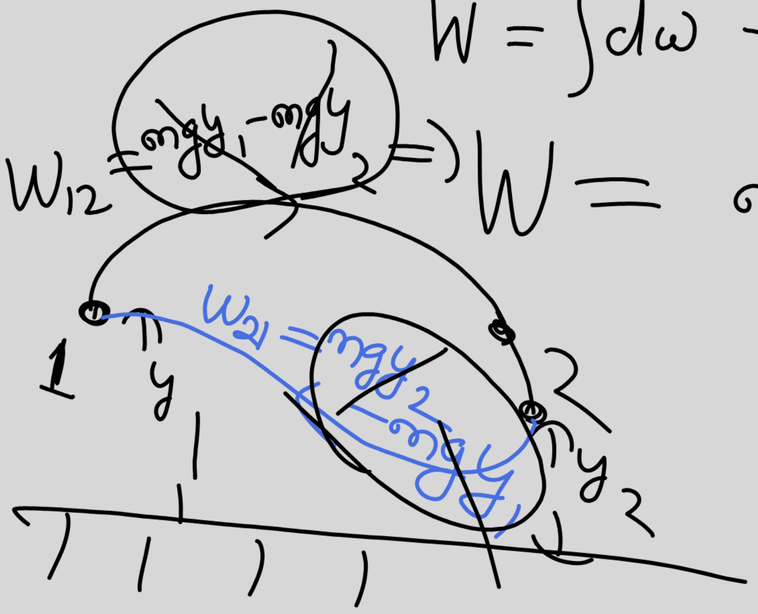
$W = \int dw = \int_{-y_1}^{-y_2} -mg\hat{j} \cdot (dx\hat{i} - dy\hat{j}) = \int_{-y_1}^{-y_2} mg\,dy$

$= mg[y]_{-y_1}^{-y_2} = mg(-y_2) - mg(-y_1) = mgy_1 - mgy_2$

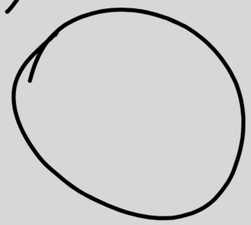


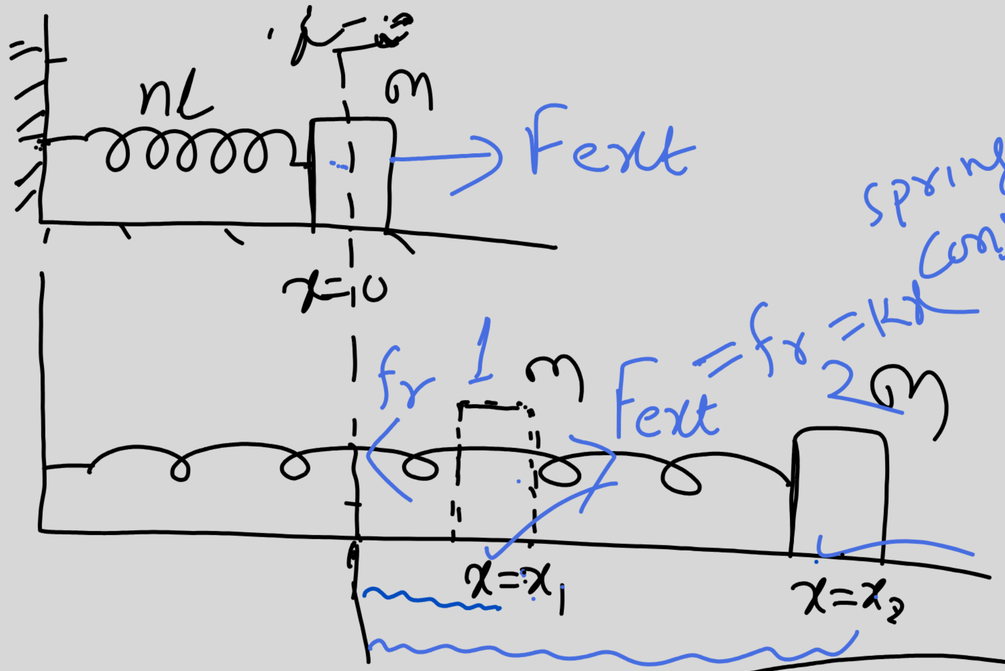
$W_{grav} = mgy_1 - mgy_2$

$= U_{grav,1} - U_{grav,2}$



$W_{12} = mgy_1 - mgy_2 \Rightarrow W = mg[y]_{-y_1}^{-y_2}$

$W_{1-2-1} = W_{12} + W_{21} =$ 



$F_r \propto -x \Rightarrow f_r = -kx$

spring const $\rightarrow |k| = \frac{f_r}{x} \rightarrow \frac{N}{m}$

$dW_{ext} = \vec{F}_{ext} \cdot d\vec{x} = F_{ext}\, dx \cos 0^\circ$

$\Rightarrow dW_{ext} = kx\,dx$

$W_{ext} = \int_{x_1}^{x_2} kx\,dx = k\left[\frac{x^2}{2}\right]_{x_1}^{x_2}$

$\Rightarrow W_{ext} = \frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2$

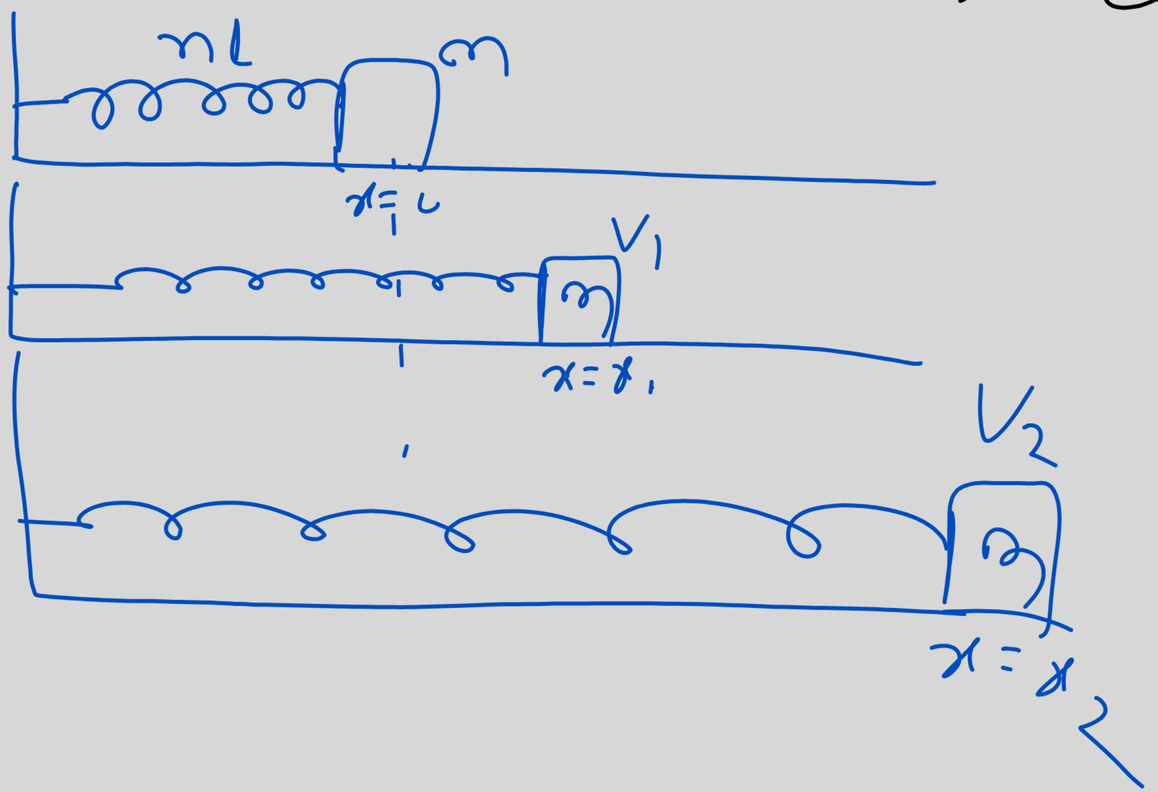
Let $x_1 = 0$, $x_2 = x$, $W_{ext} = \frac{1}{2}kx^2 = U_{el}$ (U_{grav})

$W_{ext} = U_{el,2} - U_{el,1} = \Delta U_{el}$

$W_{el} = -W_{ext} = U_{el,1} - U_{el,2}$

$W_{grav} = -W_{ext} = U_{grav,1} - U_{grav,2}$

P.t conservation of mech. Energy in presence of Grav. force & Elastic force:→



WET $W_{Total} = \Delta K = K_2 - K_1$

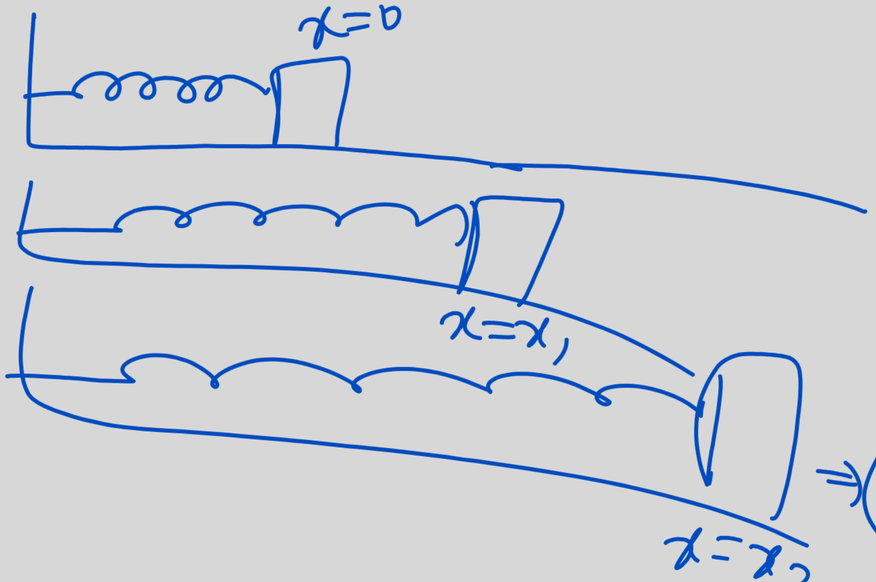
$W_{el} = K_2 - K_1$

$\Rightarrow U_{el,1} - U_{el,2} = K_2 - K_1$

$\Rightarrow U_{el,1} + K_1 = U_{el,2} + K_2$

$\Rightarrow E_1 = E_2$

Relⁿ betⁿ W_{other} with mech. Energy:→



WET $W_{Total} = \Delta K = K_2 - K_1$

$W_{grav} + W_{el} + W_{other} = K_2 - K_1$

$\Rightarrow (U_{grav,1} - U_{grav,2}) + (U_{el,1} - U_{el,2}) + W_{other} = K_2 - K_1$

$\Rightarrow (U_{grav,1} + U_{el,1} + K_1) + W_{other} = U_{grav,2} + U_{el,2} + K_2$

$(U_1 + K_1) + W_{other} = U_2 + K_2$

$\Rightarrow E_1 + W_{other} = E_2$

$\Rightarrow W_{other} = E_2 - E_1$

Exercise 7.2.

5. Evaluate the integral

$$\int \sin^2(\pi x)\cos^5(\pi x)\,dx \quad \text{, let } \pi x = y \Rightarrow \pi\,dx = dy$$

$$= \frac{1}{\pi}\int \sin^2 y \cos^5 y\,dy = \frac{1}{\pi}\int \sin^2 y \cos^4 y \cos y\,dy$$

$$= \frac{1}{\pi}\int \sin^2 y (1-\sin^2 y)^2 \cos y\,dy$$

Let $\sin y = u \Rightarrow \cos y\,dy = du$

$$= \frac{1}{\pi}\int u^2(1-u^2)^2\,du$$

$$= \frac{1}{\pi}\int u^2(1+u^4-2u^2)\,du$$

$$= \frac{1}{\pi}\left[\frac{u^3}{3}+\frac{u^7}{7}-\frac{2u^5}{5}\right]+C$$

$$= \frac{1}{\pi}\left[\frac{\sin^3\pi x}{3}+\frac{\sin^7\pi x}{7}-\frac{2}{5}\sin^5\pi x\right]+C$$

Q. 21. $\int \tan x \sec^3 x\,dx = \int \tan x \sec x \sec^2 x\,dx$

Let $u = \sec x \Rightarrow du = \sec x\tan x\,dx$

$$\int u^2\,du = \frac{u^3}{3}+C = \frac{\sec^3 x}{3}+C$$

58. Find the area of the region bounded by the given curves.

$y = \sin^3 x$, $y = \cos^3 x$, $\frac{\pi}{4} \le x \le \frac{5\pi}{4}$

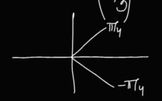
$$\text{Area} = \int_{\pi/4}^{5\pi/4} (\sin^3 x - \cos^3 x)\,dx$$

$$= \int_{\pi/4}^{5\pi/4} \sin^2 x \sin x\,dx - \int_{\pi/4}^{5\pi/4} \cos^2 x \cos x\,dx$$

$$= \int_{\pi/4}^{5\pi/4} (1-\cos^2 x)\sin x\,dx - \int_{\pi/4}^{5\pi/4} (1-\sin^2 x)\cos x\,dx$$

Let $\cos x = u \Rightarrow -\sin x\,dx = du$, $v = \sin x \Rightarrow \cos x\,dx = dv$

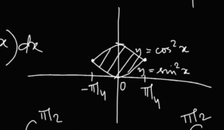
$$= -\int_{1/\sqrt{2}}^{-1/\sqrt{2}} (1-u^2)\,du - \int_{1/\sqrt{2}}^{-1/\sqrt{2}} (1-v^2)\,dv = \left(\frac{5}{3}\right)\sqrt{2}$$



HW = 3, 6, 10, 16, 36, 57

57. $y = \sin^2 x$, $y = \cos^2 x$, $-\pi/4 \le x \le \pi/4$

$$\text{Area} = \int_{-\pi/4}^{\pi/4} (\cos^2 x - \sin^2 x)\,dx$$



$$= 1$$

36. $\int_{\pi/4}^{\pi/2} \cot^3 x\,dx = \int_{\pi/4}^{\pi/2} (\cot x)(\cot^2 x)\,dx = \int_{\pi/4}^{\pi/2} (\cot x)(\operatorname{cosec}^2 x - 1)\,dx$

$$= \int_{\pi/4}^{\pi/2} \cot x \operatorname{cosec}^2 x\,dx - \int_{\pi/4}^{\pi/2} \cot x\,dx$$

4.5 Application of Congruence

Pseudorandom number

linear Congruential method

$x_{n+1} = (ax_n + b) \bmod (m)$

a = multiplier, $2 \le a < m$

b = increment, $0 \le b < m$

m = modulus

$a \bmod (b) =$

$7 \bmod (3) =$

Ex $m = 9, a = 7, b = 4, x_0 = 3$

Sol $x_{n+1} = (ax_n + b) \bmod (m)$

$x_{n+1} = (7 \cdot x_n + 4) \bmod (9)$

$\{3, 7, 8, 6, 1, 2, \dots\}$

$n = 0, x_1 = (7 \cdot x_0 + 4) \bmod (9) = (7 \cdot 3 + 4) \bmod (9) = 7$

$n = 1, x_2 = (7 \cdot x_1 + 4) \bmod (9) = (7 \cdot 7 + 4) \bmod (9) = 8$

$n = 2, x_3 = (7 \cdot x_2 + 4) \bmod (9) = (7 \cdot 8 + 4) \bmod (9) = 6$

$\dots = 1$

$\dots = 2$

Ex (a) $x_{n+1} = (4x_n) \bmod (7), x_0 = 3$

(b) $x_{n+1} = (3x_n) \bmod (11), x_0 = 2$

4.3 Primes and greatest common divisor

Prime

Ex $2, 3, 5, 7, 11, 13, \dots$ - Prime.

$4, 6, 8, 9, 10, 12, \dots$ Composite

Thm The fundamental theorem of Arithmetic.

$15 = 3 \cdot 5$

$30 = 2 \cdot 3 \cdot 5$

$17 = 17$

$\sqrt{n}$.

$n =$ int

V.I Thm

Ex show that 101 is Prime.

Sol let 101 is not Prime ✓

$\Rightarrow$ 101 is Composite number ✓

$\Rightarrow$ 101 has Prime divisor $\le \sqrt{101}$

Primes $\le \sqrt{101}$

$2, 3, 5, 7$

$2 \nmid 101, 3 \nmid 101, 5 \nmid 101, 7 \nmid 101$

$\Rightarrow$ 101 is not Composite.

$(\Leftrightarrow \Leftarrow)$ Contradiction.

$\Rightarrow$ 101 is Prime.

Greatest Common divisor (G.C.D)

g.c.d(a,b) = d

Ex a = 12, b = 24 gcd(17,23) = 1
g.c.d(12,24) = 12 gcd(24,36) = 12

$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}$
$b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n}$

$l.c.m(a,b) = p_1^{\max(a_1,b_1)} p_2^{\max(a_2,b_2)} \cdots p_n^{\max(a_n,b_n)}$

Ex a = 120, b = 500
Sol $a = 120 = 2^3 \cdot 3 \cdot 5$
$b = 500 = 2^2 \cdot 5^3 \cdot 3^0$
$g.c.d(a,b) = 2^{\min(3,2)} 3^{\min(1,0)} 5^{\min(1,3)}$
$= 2^2 \cdot 3^0 \cdot 5^1$
$= 20$

Relatively Prime. The integers a and b are relatively Prime if g.c.d(a,b) = 1.

The integers $\{a_1, a_2, \cdots a_n\}$ are Pairwise relatively Prime iff
$g.c.d(a_i, a_j) = 1, \quad 1 \le i < j \le n$

Ex g.c.d(3,7) = 1 3 and 7 are relatively Prime.
gcd(10,15) ≠ 1 10 and 15 are not " "

Ex 10, 17, 21
g.c.d(10,17) = 1
g.c.d(10,21) = 1
g.c.d(17,21) = 1
} 10, 17, 21 are Pairwise relatively Prime.

(b) 10, 15, 19
g.c.d(10,15) ≠ 1
g.c.d(15,19) = 1
g.c.d(10,19) = 1
10, 15, 19 are not Pairwise relatively Prime

least Common multiple (l.c.m)

l.c.m(10,15) = 30

Q:- All Niven Number from 1 to 1000.

```
for(int i=10; i<=1000; i++)
{
    int num=i; r, rev=0;
    while(num!=0)
    {
        r = num%10;
        rev = rev*10+r;
        num = num/10;
    }
    if(i==rev)
        s.o.pln(i);
}
```

i<=1000

1
2
3
4
5
6
7
8
9
10
12
18

n = 26

Factors :- 1, 2, 13 (n/2), ~~26~~ (n)

Count = 4.

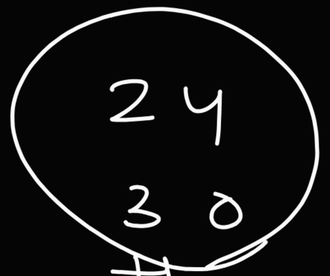
```
int n = sc.nextInt();
int Count = 2
s.o.pln("Factors are :");
s.o.pln(1);
for(int i=2; i<= n/2 ; i++)
  {
      if(n%i==0)
        {
          s.o.pln(i);
          Count++;
        }
  }
s.o.pln(n);
s.o.pln("Num of factors" : Count);
```

1 to 30 → Print nos having higest no. of factors.

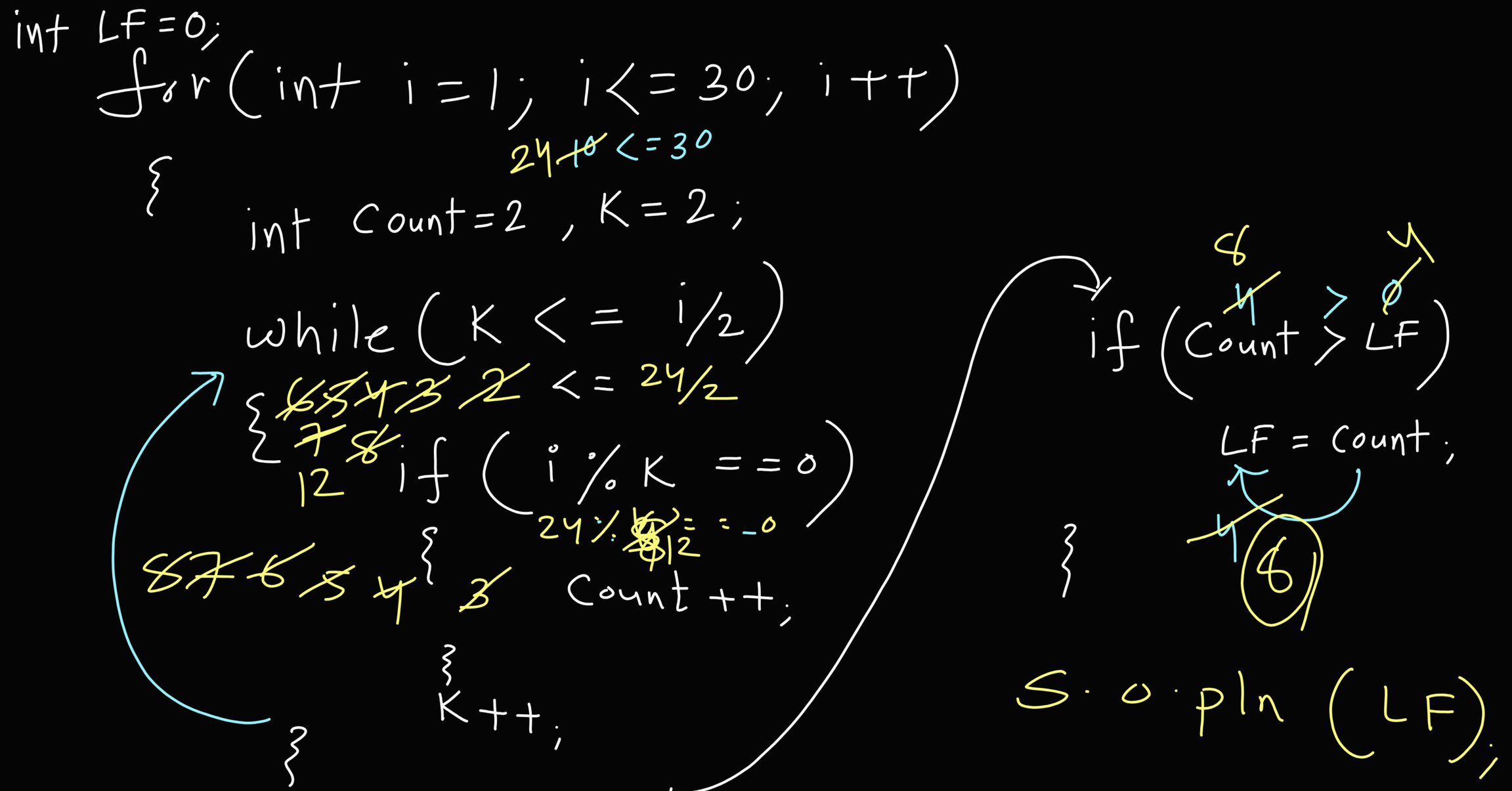
24 → 1, 2, 3, 4, 6, 8, 12, 24

30 → 1, 2, 3, 5, 6, 10, 15, 30.

O/P



Higest factors : 8


int LF=0;
for(int i=1; i<=30; i++)
{
int Count=2 , K=2;
while(K <= i/2)
{
if(i%K ==0)
Count++;
K++;
}
if(Count > LF)
LF = Count;
}
S.o.pln(LF);